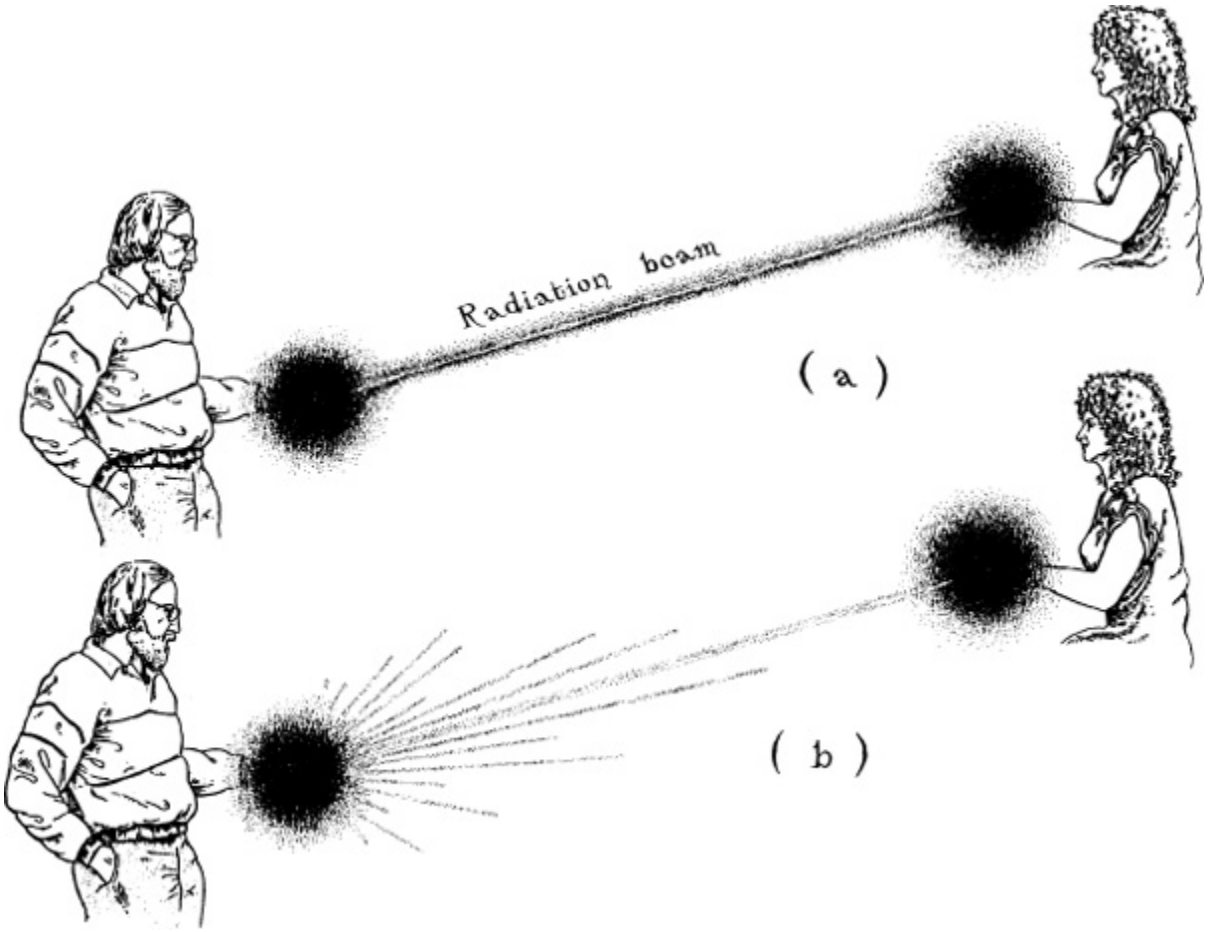


14.8 (a) The Geroch-Wald suggestion for how a wormhole might get destroyed when one tries to make it into a time machine. An intense beam of radiation zooms between the two mouths and through the wormhole, piling up on and reinforcing itself. The beam becomes infinitely energetic and destroys the wormhole. (b) What actually happens. The wormhole defocuses the beam, reducing the amount of pileup. The beam remains weak; the wormhole is not destroyed.

In this way, beginning with an arbitrarily tiny amount of radiation, a beam of infinite energy is created, coursing through space between the two wormhole mouths. As the beam passes through the wormhole, Geroch and Wald argued, it will produce infinite spacetime curvature and probably destroy the wormhole, thereby preventing the wormhole from becoming a time machine.

I drove away from Chicago and up Interstate 90 toward Madison in a daze. My mind was filled with geometric pictures of radiation beams shooting from one wormhole mouth to the other, as the mouths move toward each other. I was trying to compute, pictorially, just what *would* happen. I was trying to understand whether Geroch and Wald were right or wrong.